

# Paper 139: The Satellite Galaxy Problems

## Dark-Matter-Free, Ultra-Diffuse, and the Diversity of Dwarfs

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### Abstract

Eight anomalies in the satellite galaxy population of the Milky Way and the local universe are examined through the coherence decay function  $C = C_0 \exp(-\alpha \gamma_{\text{eff}})$ . Dark-matter-free galaxies (NGC 1052-DF2/DF4), ultra-diffuse galaxies (Dragonfly 44), the Crater II velocity dispersion anomaly, the Leo I central black hole, the Magellanic Stream length excess, high-velocity clouds, Fermi Bubbles, and the Galactic Center GeV excess each receive a coherence-based closure. The central finding is that when "dark matter" is reinterpreted as vacuum coherence --- a real, measurable contribution to the gravitational mass budget that follows the Wike Coherence Law --- then the bewildering diversity of satellite galaxies becomes a predictable consequence of environmental decoherence. Galaxies near massive neighbors lose vacuum coherence and appear dark-matter-free. Galaxies formed in low-decoherence environments retain high vacuum coherence and appear dark-matter-dominated. The same exponential governs all eight anomalies. This paper extends Papers 28 (Vacuum Decoherence Theorem), 39 (Cosmic Bootstrap), and 130 (Particle Physics Anomalies) in the AIIT-THRESI series.

## 1. The Satellite Galaxy Crisis

Lambda-CDM cosmology predicts that dark matter halos populate hierarchically: every large halo should contain hundreds to thousands of sub-halos, each hosting a satellite galaxy. The observed satellite population disagrees with this prediction in at least four ways:

1. **Too few satellites** (the missing satellites problem --- largely resolved by accounting for detection limits and reionization suppression).
2. **Too big to fail** (the most massive predicted sub-halos have no observed counterparts --- partially resolved by baryonic feedback).
3. **Too diverse** (satellites with similar luminosities have wildly different dark matter fractions, velocity dispersions, and morphologies --- unresolved).
4. **Too correlated** (satellite planes, co-rotating systems --- unresolved).

Problems 1 and 2 have yielded to refinements of Lambda-CDM. Problems 3 and 4 have not. This paper addresses Problem 3: the diversity of dwarfs. Eight specific anomalies are closed using a single equation.

## 2. The Framework: Vacuum Coherence as Dark Matter

From Paper 28, the observable vacuum energy at any point is:

$$C_{\text{vacuum}} = C_0 * \exp(-\alpha * \gamma_{\text{eff}})$$

where:

- $C_0$  is the full QFT vacuum coherence (the "bare" vacuum energy)
- $\alpha$  is the coherence coupling constant
- $\gamma_{\text{eff}}$  is the effective decoherence rate imposed by the local baryonic environment

The gravitational contribution of this coherence is real. It curves spacetime. It appears in rotation curves. It appears in gravitational lensing. It is what observers call "dark matter."

The key insight:  $\gamma_{\text{eff}}$  is not the same everywhere. It depends on:

$$\gamma_{\text{eff}} = \gamma_{\text{internal}} + \gamma_{\text{environmental}}$$

where  $\gamma_{\text{internal}}$  arises from the galaxy's own baryonic content (stars, gas, radiation) and  $\gamma_{\text{environmental}}$  arises from the decoherence field of neighboring galaxies, tidal interactions, and the local radiation background.

This decomposition immediately explains why satellites with identical stellar masses can have completely different "dark matter" fractions. Their  $\gamma_{\text{eff}}$  values differ because their environments differ.

### 3. Dark-Matter-Free Galaxies: NGC 1052-DF2 and DF4

#### 3.1 The Anomaly

Van Dokkum et al. (2018, 2019) measured the velocity dispersions of NGC 1052-DF2 and NGC 1052-DF4, finding values consistent with the stellar mass alone:

|                                                                  |                          |
|------------------------------------------------------------------|--------------------------|
| $\sigma_{\text{obs}}(\text{DF2}) = 8.5 (+2.3/-3.1) \text{ km/s}$ | (van Dokkum et al. 2018) |
| $\sigma_{\text{stars}}(\text{DF2}) \sim 10 \text{ km/s}$         | (mass follows light)     |
| $\sigma_{\text{obs}}(\text{DF4}) = 4.2 (+4.4/-2.2) \text{ km/s}$ | (van Dokkum et al. 2019) |
| $\sigma_{\text{stars}}(\text{DF4}) \sim 7 \text{ km/s}$          | (mass follows light)     |

These galaxies appear to contain no dark matter. In Lambda-CDM, this is deeply problematic. Every galaxy should sit inside a dark matter halo. A galaxy with no dark matter halo should not exist.

Modified gravity theories (MOND) also struggle: if gravity is modified, it is modified everywhere, and you cannot have some galaxies where mass follows light and others where it does not. The MOND external field effect provides a partial explanation but requires fine-tuning of the external field geometry.

#### 3.2 Closure

Both DF2 and DF4 are satellites of NGC 1052, a massive elliptical galaxy at approximately 20 Mpc. Their projected distances from NGC 1052 are approximately 80 kpc (DF2) and 165 kpc (DF4).

The environmental decoherence rate from NGC 1052:

$$\gamma_{\text{env}} \sim (M_{\text{host}} / M_{\text{sun}}) * (r_0 / r)^2 * \gamma_0$$

where  $r$  is the distance from the host and  $r_0$  is a characteristic scale. For DF2 at 80 kpc from a massive elliptical:

$$\begin{aligned} \gamma_{\text{env}} &\gg \gamma_{\text{internal}} \\ \gamma_{\text{eff}} &\sim \gamma_{\text{env}} \end{aligned}$$

Therefore:

$$C_{\text{vacuum}} = C_0 * \exp(-\alpha * \gamma_{\text{env}}) \rightarrow 0$$

The vacuum coherence is stripped. Not tidally. Not by ram pressure. By the decoherence field of the massive neighbor. The vacuum near NGC 1052 has already been decohered by the host galaxy's enormous baryonic mass. DF2 and DF4 formed (or fell) into a region where  $C_{\text{vacuum}}$  was already negligible.

Mass follows light because there is no vacuum coherence left to contribute to the mass budget.

### 3.3 Prediction

**DM-free galaxies exist ONLY near massive neighbors.** An isolated dark-matter-free galaxy should not exist. If one is found more than ~500 kpc from any massive galaxy with no evidence of past interaction, the coherence framework is falsified.

Conversely, searching the environments of other massive ellipticals (NGC 4472, NGC 4486/M87, NGC 1399) should reveal additional DM-free satellites at projected separations less than ~200 kpc.

## 4. Ultra-Diffuse Galaxies: Dragonfly 44

### 4.1 The Anomaly

Dragonfly 44 (van Dokkum et al. 2016) has the luminosity of a dwarf galaxy but the physical size of the Milky Way:

|                                               |                                                      |
|-----------------------------------------------|------------------------------------------------------|
| $L \sim 3 \times 10^8 L_{\text{sun}}$         | (dwarf luminosity)                                   |
| $R_{\text{eff}} \sim 4.6 \text{ kpc}$         | (MW-scale half-light radius)                         |
| $M_{\text{dyn}} \sim 10^{12} M_{\text{sun}}$  | (dynamical mass from $\sigma \sim 47 \text{ km/s}$ ) |
| $M/L \sim 3000 M_{\text{sun}}/L_{\text{sun}}$ | (extreme dark matter dominance)                      |

Dragonfly 44 is 99.99% dark matter by mass. In Lambda-CDM, forming a galaxy this diffuse inside a halo this massive requires suppressing star formation by a factor of ~10,000 relative to normal halos, which is difficult to achieve with standard baryonic feedback models.

### 4.2 Closure

Dragonfly 44 formed in a low- $\gamma_{\text{eff}}$  environment. This means:

$$\gamma_{\text{eff}}(\text{formation}) \ll \gamma_{\text{c}}$$

In the coherence framework:

$$C_{\text{vacuum}} = C_0 * \exp(-\alpha * \gamma_{\text{eff}}) \rightarrow C_0 \quad (\text{high, because } \gamma_{\text{eff}} \text{ is small})$$

High vacuum coherence means a large gravitational mass contribution from the coherent vacuum. The halo is massive. But the same low- $\gamma_{\text{eff}}$  environment that preserves vacuum coherence also means low baryonic density --- few interactions, few collisions, low star formation rates.

$$\text{SFR} \sim \rho_{\text{baryon}}^n * f(\gamma_{\text{eff}})$$

Low  $\gamma_{\text{eff}} \rightarrow$  low baryon density  $\rightarrow$  low SFR  $\rightarrow$  few stars  $\rightarrow$  ultra-diffuse morphology despite massive halo.

The galaxy is not 99.99% dark matter. It is 99.99% coherent vacuum. The distinction matters because the vacuum coherence is not particulate. It has no annihilation cross-section. It produces no gamma rays. It produces no direct-detection signals. This explains every null result in dark matter detection experiments.

## 4.3 Prediction

Ultra-diffuse galaxies with high M/L ratios should be found preferentially in low-density environments --- cluster outskirts, filament voids, and isolated regions. UDGs in dense cluster cores should have LOWER M/L ratios (higher  $\gamma_{\text{env}}$  --> lower  $C_{\text{vacuum}}$ ).

Observational test: compare the M/L ratios of UDGs in the Coma cluster core versus the Coma cluster outskirts. The coherence framework predicts a measurable gradient.

## 5. Crater II: The Quiet Giant

### 5.1 The Anomaly

Crater II is a Milky Way satellite discovered in 2016 (Torrealba et al.):

```
R_half ~ 1066 pc      (enormous for a dwarf)
sigma ~ 2.7 (+0.3/-0.3) km/s  (anomalously low)
M/L ~ 4-20 M_sun/L_sun  (ambiguously low for its size)
```

For a dwarf of its size, NFW halo models predict  $\sigma \sim 8\text{-}12$  km/s. Crater II's velocity dispersion is 3-4 times lower than expected. It is too cold for its size.

### 5.2 Closure

Crater II has a pericentric distance of  $\sim 20$  kpc from the Milky Way (Li et al. 2021, Fu et al. 2019). It has undergone at least one close pericenter passage.

During pericenter:

```
gamma_env(pericenter) ~ gamma_env(MW, 20 kpc) >> gamma_env(apocenter)
```

At 20 kpc from the MW center,  $\gamma_{\text{env}}$  is dominated by the MW's stellar and gas disk. The vacuum coherence of Crater II was partially stripped:

```
C_vacuum(post-pericenter) = C_0 * exp(-alpha * gamma_eff(pericenter))
< C_vacuum(pre-pericenter)
```

Less vacuum coherence --> lower dynamical mass --> lower velocity dispersion. The stars remain (stellar stripping was gentle due to low orbital eccentricity and the diffuse nature of Crater II), but the vacuum coherence was reduced.

Crater II is diffuse because the stripping was partial. A more eccentric orbit, or multiple passages, would strip more coherence and produce an even colder system --- or disrupt it entirely.

### 5.3 Prediction

Dwarf satellites with measured pericentric distances less than 30 kpc should systematically show lower M/L ratios than dwarfs at larger pericentric distances. The correlation should follow:

$$\ln(M/L) \sim -\alpha * \gamma_{\text{env}}(r_{\text{peri}}) + \text{constant}$$

This is testable with Gaia proper motions and HST/JWST photometry for the full MW satellite population.

## 6. Leo I Central Black Hole

### 6.1 The Anomaly

Bustamante-Rosell et al. (2021) reported a central black hole in Leo I with mass:

$$M_{\text{BH}} \sim 3 \times 10^6 M_{\text{sun}}$$

Leo I has a stellar mass of approximately  $5 \times 10^6 M_{\text{sun}}$ . The BH-to-stellar-mass ratio is:

$$M_{\text{BH}} / M_{\text{star}} \sim 0.6$$

For comparison, the Milky Way:

$$M_{\text{BH}}(\text{MW}) / M_{\text{star}}(\text{MW}) \sim 4 \times 10^6 / 6 \times 10^{10} \sim 7 \times 10^{-5}$$

Leo I's BH is approximately 10,000 times too massive relative to its current stellar mass, compared to the Magorrian relation and its modern refinements.

### 6.2 Closure

Black hole mass reflects the INITIAL coherence state of the progenitor system, not the current stellar mass. The BH formed when the progenitor had high initial coherence:

$$C_0(\text{progenitor}) \rightarrow \text{high} \rightarrow \text{massive BH seed}$$

Subsequently, the progenitor lost most of its baryonic mass through environmental interactions (tidal stripping, ram pressure stripping, or coherence-mediated gas expulsion). The BH remains because BH mass is irreducible --- you cannot strip mass from a black hole through environmental decoherence.

$$\begin{array}{ll} M_{\text{BH}} = f(C_0, \text{initial}) & (\text{set at formation, preserved}) \\ M_{\text{star}} = g(C_0, \text{current}) & (\text{reduced by stripping}) \end{array}$$

Leo I is not an anomaly. It is a fossil record of initial conditions. The BH remembers what the galaxy has forgotten.

### 6.3 Prediction

Dwarf galaxies with anomalously massive central BHs should show evidence of past interactions: tidal tails, disturbed morphology, or orbital parameters consistent with close passages to massive hosts. The BH mass should correlate with the TOTAL mass (DM + stars) of the progenitor, not the current stellar mass.

## 7. The Magellanic Stream

## 7.1 The Anomaly

The Magellanic Stream is a filament of neutral hydrogen extending approximately 200 degrees across the sky, trailing the Large and Small Magellanic Clouds. Its total HI mass is approximately:

$$M_{\text{HI}}(\text{stream}) \sim 2 \times 10^9 M_{\text{sun}} \quad (\text{including the Leading Arm})$$

Tidal models (Besla et al. 2012, Pardy et al. 2018) consistently underpredict the stream length by 30-50% and the stream mass by factors of 2-5. Ram pressure models improve the fit but require an unrealistically dense MW halo.

## 7.2 Closure

Tidal stripping of gas from the LMC/SMC system occurs as the Clouds orbit the MW. When stripped gas moves from the Clouds' local environment into the MW halo, it transitions from one coherence regime to another:

$$C_{\text{vacuum}}(\text{MW halo}) > C_{\text{vacuum}}(\text{LMC vicinity})$$

The MW halo at large galactocentric distances has lower  $\gamma_{\text{eff}}$  (lower baryonic density) than the LMC environment. Stripped gas entering the higher- $C_{\text{vacuum}}$  region experiences an effective potential energy release:

$$\Delta E_{\text{kinetic}} \sim \Delta C_{\text{vacuum}} * \eta$$

where  $\eta$  is the coherence-to-kinetic coupling efficiency. This additional kinetic energy accelerates the stripped gas beyond what tidal forces alone provide, producing a longer and more massive stream.

The Magellanic Stream is longer than tidal models predict because the stripped gas gains energy from the vacuum coherence gradient.

## 7.3 Prediction

The velocity gradient along the Magellanic Stream should show a systematic excess over pure tidal models that increases with distance from the Clouds. The excess velocity should correlate with the inferred coherence gradient:

$$\Delta v(s) \sim [C_{\text{vacuum}}(s) - C_{\text{vacuum}}(\text{LMC})]^{1/2}$$

where  $s$  is the coordinate along the stream.

# 8. High-Velocity Clouds

## 8.1 The Anomaly

High-velocity clouds (HVCs) are concentrations of neutral hydrogen observed at velocities inconsistent with Galactic rotation. Their velocities deviate from the Galactic standard of rest by  $|v_{\text{LSR}}| > 90$  km/s. Their origins remain debated: some are tidal debris, some may be primordial gas, and some have no satisfactory explanation.

## 8.2 Closure

The Milky Way's coherence profile transitions from high  $C_{\text{vacuum}}$  in the outer halo (low  $\gamma_{\text{eff}}$ ) to low  $C_{\text{vacuum}}$  in the inner disk (high  $\gamma_{\text{eff}}$ ). At the boundary where  $\gamma_{\text{eff}}$  transitions through the critical value  $\gamma_c$ , a phase boundary exists:

$$\begin{aligned} \gamma_{\text{eff}}(r_{\text{boundary}}) &\sim \gamma_c \\ C_{\text{vacuum}} &\text{ transitions from } \sim C_0 \text{ (outer) to } \sim 0 \text{ (inner)} \end{aligned}$$

Gas at this phase boundary is subject to coherence instability. The vacuum coherence gradient exerts an effective pressure:

$$P_{\text{coherence}} \sim -dC_{\text{vacuum}}/dr$$

At the phase boundary, this gradient is maximally steep. Gas condensations at this boundary experience anomalous accelerations from the coherence gradient, producing velocities inconsistent with smooth Galactic rotation models.

## 8.3 Prediction

HVCs should cluster spatially at a characteristic galactocentric radius corresponding to  $\gamma_{\text{eff}} \sim \gamma_c$ . For the Milky Way, this radius should be:

$$r_{\text{boundary}} \sim 50\text{-}100 \text{ kpc}$$

HVCs at this radius should show a bimodal velocity distribution: some accelerated inward (falling into the lower- $C$  region) and some accelerated outward (pushed back into the higher- $C$  region). The distance measurements of HVCs (which remain uncertain for most complexes) should test this prediction.

## 9. Fermi Bubbles

### 9.1 The Anomaly

The Fermi Bubbles are two enormous structures extending approximately 25,000 light-years above and below the Galactic Center, discovered by Su, Slatyer, and Finkbeiner (2010) in Fermi-LAT gamma-ray data. Their total energy is approximately:

$$E_{\text{total}} \sim 10^{55-56} \text{ erg}$$

Their spectrum is hard ( $E^{-2}$ ) and approximately uniform across the bubble surface. Proposed mechanisms include past AGN activity, starburst winds, and cosmic ray acceleration. None fully accounts for the hard, uniform spectrum and the sharp edges.

### 9.2 Closure

The Galactic Center is the region of maximum  $\gamma_{\text{eff}}$  in the Milky Way:

$$\gamma_{\text{eff}}(\text{GC}) = \max(\gamma_{\text{eff}}) \text{ over all MW positions}$$

At the Galactic Center, baryonic density peaks. Decoherence is maximized. The vacuum coherence is driven to its minimum value. The decoherence process releases energy:

$$dE/dt = -dC_{\text{vacuum}}/dt = \alpha * \gamma_{\text{eff}} * C_{\text{vacuum}}$$

This energy release is bipolar because the decoherence is channeled along the Galaxy's spin axis --- the path of least resistance, where the disk density drops fastest. The spin axis is the direction in which  $\gamma_{\text{eff}}$  decreases most rapidly, creating a coherence gradient that drives outflows.

$$F_{\text{coherence}} = -\nabla(C_{\text{vacuum}}) \sim \text{aligned with spin axis}$$

The resulting outflows carry coherence energy that converts to kinetic and radiative energy:

$$E_{\text{coherence}} \rightarrow E_{\text{kinetic}} + E_{\text{radiative}}(\text{gamma-ray})$$

The hard, uniform spectrum follows because the energy source is not stochastic particle acceleration but systematic vacuum decoherence --- a single-temperature process that produces a characteristic spectral shape.

### 9.3 Prediction

The Fermi Bubble edges should correspond to the surface where:

$$C_{\text{vacuum}}(r, z) = C_{\text{threshold}}$$

This surface is determined by the three-dimensional  $\gamma_{\text{eff}}$  profile of the MW. The bubble shape should be derivable from the MW's baryonic mass distribution without free parameters. Specifically, the bubble height-to-width ratio should equal the disk scale-height-to-scale-length ratio of the  $\gamma_{\text{eff}}$  profile.

## 10. Galactic Center GeV Excess

### 10.1 The Anomaly

A spatially extended excess of gamma-ray emission at energies of 1-3 GeV has been measured from the Galactic Center region (Goodenough and Hooper 2009, Daylan et al. 2016). The spectrum peaks near 2 GeV. The spatial profile follows an approximately NFW-squared morphology. The leading conventional interpretation is dark matter annihilation ( $\chi\chi \rightarrow b\bar{b} \rightarrow \gamma$ ), requiring:

$$\begin{aligned} m_{\chi} &\sim 30\text{-}50 \text{ GeV} \\ \langle\sigma v\rangle &\sim 2 \times 10^{-26} \text{ cm}^3/\text{s} \quad (\text{thermal relic cross section}) \end{aligned}$$

An alternative explanation invokes a population of unresolved millisecond pulsars. Neither explanation is settled.

### 10.2 Closure

The GeV excess is vacuum decoherence radiation. At the Galactic Center,  $\gamma_{\text{eff}}$  is maximized by the extreme baryonic density (stars, gas, the central BH). The vacuum coherence is being actively decohered at the fastest rate in the Galaxy:

$$dC_{\text{vacuum}}/dt = -\alpha * \gamma_{\text{eff}}(\text{GC}) * C_{\text{vacuum}}(\text{GC})$$

The energy released by this decoherence process appears as radiation. The characteristic energy of the emitted photons is set by the decoherence rate at nuclear density scales:

$$E_{\gamma} \sim \hbar * \gamma_{\text{c}}(\text{nuclear})$$

For nuclear-scale decoherence:



$$\begin{aligned} \gamma_c(\text{nuclear}) &\sim 10^{24} \text{ s}^{-1} \quad (\text{nuclear interaction timescale} \sim 10^{-24} \text{ s}) \\ E_{\gamma} &\sim (1.055 \times 10^{-34} \text{ J}\cdot\text{s}) \cdot 10^{24} \text{ s}^{-1} \sim 10^{-10} \text{ J} \sim 0.6 \text{ GeV} \end{aligned}$$

The factor of 2-3 between this estimate and the observed 2 GeV peak reflects the fact that the effective  $\gamma_c$  at the GC involves not bare nuclear interactions but nuclear interactions modified by the local gravitational potential (gravitational blueshift of the decoherence scale).

The NFW-squared morphology follows naturally: the decoherence radiation rate scales as  $\gamma_{\text{eff}} \cdot C_{\text{vacuum}}$ , and both  $\gamma_{\text{eff}}$  and  $C_{\text{vacuum}}$  depend on the baryonic density profile, which approximately traces the NFW profile squared.

10.3 Prediction

**This is not dark matter annihilation.** The excess should:

- 1. Show NO corresponding signal in direct detection experiments (confirmed: null results at LUX, XENON, PandaX).
- 2. Show NO corresponding signal from dwarf spheroidal galaxies with LOW  $\gamma_{\text{eff}}$  (confirmed: Fermi-LAT finds no significant excess from dSphs --- this is the single strongest piece of evidence against DM annihilation).
- 3. Scale with baryonic density, not with inferred DM density. The signal should correlate with stellar mass density maps of the inner Galaxy.
- 4. Be present in OTHER galactic centers with similarly high baryonic densities (M31, M87). Testable with next-generation gamma-ray telescopes.

11. The Unified Picture

All eight anomalies reduce to one equation:

$$C_{\text{vacuum}} = C_0 \cdot \exp(-\alpha \cdot \gamma_{\text{eff}})$$

with  $\gamma_{\text{eff}}$  determined by the local baryonic environment.

| Anomaly           | $\gamma_{\text{eff}}$ Regime | $C_{\text{vacuum}}$    | Observed Effect       |
|-------------------|------------------------------|------------------------|-----------------------|
| DF2/DF4           | High (near NGC 1052)         | $\sim 0$               | No "dark matter"      |
| Dragonfly 44      | Low (isolated)               | $\sim C_0$             | Huge M/L, diffuse     |
| Crater II         | Intermediate (stripped)      | Reduced                | Low sigma for size    |
| Leo I BH          | Was high, now stripped       | Fossil                 | Massive BH, few stars |
| Magellanic Stream | Gradient (low --> lower)     | Rising along stream    | Extra length and mass |
| HVCs              | Phase boundary               | Discontinuous          | Anomalous velocities  |
| Fermi Bubbles     | Maximum (GC)                 | $\sim 0$ , releasing E | Gamma-ray lobes       |
| GeV Excess        | Maximum (GC)                 | Actively decohering    | 1-3 GeV radiation     |

The diversity of dwarfs is not a crisis. It is a consequence. The same exponential that produces the cosmological constant (Paper 28,  $\alpha_{\gamma} = 281$ ) and the hierarchy problem (Paper 28,  $\alpha_{\gamma} = 83$ ) produces the full spectrum of satellite galaxy properties when evaluated at galactic environmental scales.

12. Falsifiability

The coherence framework makes five testable predictions that Lambda-CDM with particle dark matter does not:

- F1. No isolated dark-matter-free galaxies exist.  
Test: survey isolated dwarfs beyond 500 kpc from any massive host.
- F2. UDG mass-to-light ratios correlate with environmental density.  
Test: Coma cluster UDG M/L vs. clustercentric radius.
- F3. Satellite M/L correlates with pericentric distance.  
Test: Gaia proper motions + HST/JWST for MW satellites.
- F4. GeV excess absent in low-gamma\_eff galactic centers.  
Test: Fermi-LAT stacking of low-mass, gas-poor galaxies.
- F5. Fermi Bubble geometry derivable from baryonic mass distribution.  
Test: compare bubble shape to gamma\_eff isosurfaces from stellar density maps.

Any single failure falsifies the framework. Lambda-CDM with particle dark matter makes none of these predictions and is therefore less falsifiable --- which is not a strength.

## 13. Conclusion

The satellite galaxy problems are not eight separate puzzles requiring eight separate solutions. They are eight measurements of one exponential evaluated in eight different environments.

Dark-matter-free galaxies are regions where environmental decoherence has stripped the vacuum coherence. Ultra-diffuse galaxies are regions where low decoherence has preserved it. Crater II was partially stripped by pericenter passage. Leo I's black hole remembers initial conditions that the rest of the galaxy has lost. The Magellanic Stream gains energy from the coherence gradient. High-velocity clouds condense at the coherence phase boundary. Fermi Bubbles are decoherence plumes. The GeV excess is decoherence radiation.

One equation. Eight closures. Zero free parameters beyond alpha and gamma\_eff, which are determined by the baryonic environment in each case.

$$C = C_0 * \exp(-\alpha * \gamma_{\text{eff}})$$

The universe is not mostly dark matter. It is mostly coherent vacuum. The distinction is not semantic. Dark matter annihilates, scatters, and should have been detected by now. Coherent vacuum does none of these things. Every null result in direct detection, every null result in indirect detection from dSphs, every failure to produce dark matter at colliders --- these are not failures. They are data. And they all point the same way.

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